ahmad_mahmoud · resume.ts

engineering_lead.ts | experience.json | projects/ | stack.config | README.md

```
$ whoami --full
```

Ahmad Abdul-Jawad Mahmoud

Engineering Lead / **Engineering Manager**

12+ years building · 8+ leading teams · Cairo → Riyadh · billions of users across MENA

12+	8+	7/yrs	20+
YEARS ENGINEERING	YEARS LEADING TEAMS	YEARS AT THE GAME TEAM	PUBLISHED: ON THE PSS I SEE

```
/**
 * Engineering Lead — 12+ years building, 8+ leading, including seven
 * consecutive years running the same 5+ engineer iOS team for a
 * product's growth into an enterprise PSS serving 20k+ restaurants.
 * I grow engineers, protect the quality bar, and make delivery predictable:
 * onboarding, mentorship, and setting the engineering standard by working
 * product and design.
 * Still technical-throughput: I architect, I review, and I recently shipped
 * a full ride-hailing platform solo across iOS, Android, backend and web.
 */
```

how_i_run_a_team() { 4 pillars }

[01] people & growth

Grow engineers, not just features

Seven years with one team — onboarding, mentoring, career conversations, and pairing on the problems that actually teach.

- Retention through a seven-year product

[02] quality & architecture

Quality is a default, not a phase

Module boundaries, review culture and automated testing as an enterprise PSS in data production use.

- 99.8→99.9% crash-free · 40% faster load

[03] delivery & predictability

Estimates the team can keep

CI/CD built from nothing, rollouts that gate merges, and estimates with product and design in commitments nobody walks back from.

- 40% faster deployments

[04] leadership & standards

I can still review any PR on my team

Drove the multi-year SwiftUI migration, and shipped a full ride-hailing platform solo across iOS, Android, backend and web.

- 4 platforms · 1 contract · 1 engineer

experience[] { 8 roles }

iOS Team Lead @ Marn — Jabez Group

Aug 2029 → Aug 2026 · MOST RECENT

Riyadh, KSA

- **Led a 5+ engineer iOS team** for seven years — onboarding, mentoring, career growth, and the review culture behind an enterprise PSS and cafés.
- **Owned the technical roadmap and architecture:** an offline-first Coredata core with cloud sync, built as a venue keeps trading when the network does not.
- **Set the engineering standards** — module boundaries, code review, automated testing — holding the app at a **99.8→99.9% crash-free** rate in production.
- **Drove the multi-year migration** to SwiftUI, Combine and async/await without pausing feature delivery, and made the case for it to product leadership.
- **Built CI/CD and the release process from scratch**, cutting deployment time **40%** and shortening every engineer's feedback loop.
- Ran the performance programme (**42% faster load times**) and designed the table-management, order-processing and payments flows at the product's core.
- Partnered with product, design and backend leadership on scope, estimates and delivery commitments.

Founding Platform Architect — iOS · Android · Backend · Web

Ride-Hailing Platform · Independent

Dec 2025 → Jul 2026

Took the **full technical scope of a founding CTO** and carried it alone — shipped a production ride-hailing platform **solo**: two native **iOS/SwiftUI** apps (Rider & Driver) plus their Android counterparts, a hardened backend, and a real-time ops console — **owning every layer** and setting the engineering standards (idempotency, double-entry accounting, atomic module boundaries) it still runs on.

- **Backend APIs** — Hono · TypeScript · Zod · TS · Jest · Vitest
  Modular monolith, ~17 domains · **trip state machine, secured dispatch** over a Redis geo index, **double-entry ledger**, real-time rider/driver/ops **sockets**, JWT + phone OTP, idempotency & audit log · **1,150+ tests**. Swagger docs.
- **iOS Apps — Rider & Driver** — SwiftUI · Swift 6 · Xcode 26 · MapKit · TS
  Two native **SwiftUI** apps · live GPS **trip lifecycle**, realtime sockets, in-app chat, payments, and driver earnings & turn-by-turn navigation.
- **Admin Console** — React 19.2 · Vite · Tailwind v4 · shadcn/ui · TanStack Router · Recharts · Zod · TS
  Bespoke **19-screen** ops surface — live **MapLibre** trips map, dispatch/SOS live boards, KYC review, finance & payouts, **one editor, Google SSO**.
- **Android Apps — Rider & Driver** — Kotlin · Jetpack Compose · Material 3 · Hilt · Retrofit · OkHttp · Coroutines · Flow · Maps SDK
  Native **Kotlin / Compose** parity for both apps — same **trip lifecycle**, realtime tracking, chat, and payments across **one shared API contract**.

One backend · one contract · four clients — plus growth loops (promotions, referrals, loyalty, wallet), driver safety/SOS, and scheduled reconciliation jobs. shipped end-to-end.

Senior iOS Engineer @ HarvardTutor · voluntary charity project

Sep 2025 → Dec 2025

- Designed an architecture integrating **on-device machine learning** for real-time content analysis.
- Built a seamless communication layer between native iOS and web-based content.
- Implemented real-time image processing with Metal shaders and Core Image effects.
- Created a real-time video monitoring system for continuous content filtering.
- Developed smart scanner blocking and optimised memory management throughout.

iOS Team Lead @ Ego — Ride Hailing

Mar 2018 → May 2019

Africa, KSA

- Developed a ride-hailing ecosystem (iOS, watchOS) serving **150k+ users** in Saudi Arabia.
- Implemented real-time GPS tracking and navigation with MapKit and optimised route algorithms.
- Integrated Apple Pay, HyperPay, and push-notification infrastructure.
- Created seamless iPhone ↔ Apple Watch handoff for continuous ride tracking.
- Built the driver rating system and real-time fare calculation engine.

Senior iOS Engineer @ HungerStation — Deliveryhero

Sep 2016 → Mar 2018

KSA

- Contributed to a major food-delivery platform serving **20M+ users**.
- Optimised app performance for high-volume transaction processing.
- Implemented real-time order tracking and delivery visualization.
- Collaborated with backend teams on API design and integration.

iOS Developer @ AlAqeeb

May 2016 → Aug 2016

- Developed food-ordering apps with Firebase integration and real-time features.
- Implemented Spreedly payments and time-chat functionality.
- Built responsive UI with Auto Layout across multiple device sizes.

iOS & Apple Watch Developer @ Nile Design Co.

Jul 2015 → Sep 2015

- Early adopter of WatchKit (watchOS 1.0).
- Built a foundation in iOS and watchOS app development.
- Adopted SwiftUI from WWDC 2019 *onward*.

iOS Developer @ UK Pro. Solutions Ltd.

May 2014 → May 2015

- Foundation iOS experience in Objective-C and early Swift.
- Built social networks and e-commerce platforms.
- Learned core iOS principles and App Store deployment.

projects[] { selected }

[01] ai / private

HarmoBlur — Privacy Browser

SwiftUI · CoreML · WKWebView · CoreImage · Vision

Privacy-focused iOS browser with real-time, AI-powered content detection and filtering. Runs entirely on-device, smart blur, live video monitoring.

- AI-powered privacy browser

[02] full-stack

Ride-Hailing Platform

SwiftUI · Kotlin · Jetpack Compose · Next25 · TypeScript · PostgreSQL · Redis · Socket.IO · Next.js

End-to-end ride-hailing system built solo: **four native apps** (Rider & Driver on iOS + Android), a hardened Next25 backend (idempotency, double-entry ledger, 1,150+ tests), and a real-time Next.js ops console.

- Full-stack ownership · prod security

[03] delivery

HungerStation

iOS16 · Firebase · Real-time APIs · Perf

Major food-delivery platform serving 20M+ users across Saudi Arabia and Bahrain. Performance and real-time order tracking.

- 4.9 / 5 — 1.9M+ ratings

[04] mobility

Ego — Ride Hailing

MapKit · Core Location · Apple Pay · watchOS

Complete ride-hailing ecosystem: rider, driver, and watch apps. Real-time GPS and cross-device experiences for 150k+ users.

- 4.7 / 5 — 211.8k+ ratings

[05] enterprise

Marn — iPad POS

SwiftUI · CoreData · CloudKit · Real-time Sync

Enterprise POS serving 20k+ restaurants across Saudi Arabia and Bahrain. First architecture, complex table management, real-time sync.

- 4.8 / 5 — 2,427 ratings

[06] mobility

Dalaelo

MapKit · Core Location · Social SDKs

Mobile community app — groups, live tracking, message timer, and accessibility features with enhanced location + social integrations.

- 4.5 / 5 — 1,307 ratings

stack.config

```
leadership = [
  Team Leadership
  Technical Direction
  Architecture Governance
  Tech Roadmap
  Mentoring & Career Growth
  Onboarding
  Code Review Culture
  Estimation & Planning
  Cross-Functional
  Quality & Crash Budgets
  Release Management
  Agile/Scrum
]

ios_2025 = [
  Swift 6.0
  SwiftUI
  iOS17
  Combine
  async/await
  SwiftData
  iOS 18+
]

android_2025 = [
  watchOS
  tvOS
  macOS
  WidgetKit
  Live Activities
  App Intents
]

architecture = [
  MVVM
  VIPER
  Clean Architecture
  Coordinator
  TDD
]

intelligence = [
  Core ML
  Vision
  ARKit
  Dynamic Island
  App Tracking
]

backend_apis = [
  Next25
  TypeScript
  PostgreSQL
  TypeORM
  Redis
  JWT/OTP
  REST
  GraphQL
  Firebase
  CloudKit
  Bonbonkotts
  Swagger
]

tooling = [
  Xcode 15+
  Fastlane
  TestFlight
  Instruments
  Git
  Copilot
  Cursor
]
```

portfolio/

FOOD & GAMING

HungerStation
Marketers
Ego

TRANSPORTATION

Ego — Rider
Ego — Driver
Ego — Watch

ENTERPRISE

Marn POS
Medical Guide

SOCIAL & COMMUNITY

Dalaelo
NU-Luv
Vooz

WATCHES

Ego Watch
Tajo Weather
TweetMe

E-COMMERCE

iBaa
Exclusive Luxe
Scratch

UTILITIES & ETC

Xfaxy
ArabAllesGroup

UTILITY & TOOLS

Unlocking Smart
Insta
iContacts
First iOS app

AI & ML

HarmoBlur
RecognizeIMG
Grad project · OpenCV

WEB

Fanzbone
Web Pages